

[Linux 50 Commands for SOC L1 Analyst]

These commands are very useful for log analysis, troubleshooting, incident investigation, monitoring, networking, and basic threat hunting.

[1. File & Directory Operations]

1. pwd

```
pwd
```

Meaning: Present Working Directory (Shows current directory).

2. ls

```
ls  
ls -la
```

Meaning: List files and folders.

3. cd

```
cd /var/log
```

Meaning: Change directory.

4. mkdir

```
mkdir logs
```

Meaning: Create new directory.

5. rm

```
rm test.txt  
rm -rf folder
```

Meaning: Remove file or folder.

6. cp

```
cp file1.txt backup.txt
```

Meaning: Copy files.

7. mv

```
mv old.txt new.txt
```

Meaning: Move or rename files.

8. touch

```
touch notes.txt
```

Meaning: Create empty file.

[2. Text Manipulation & Log Searching]

9. cat

```
cat auth.log
```

Meaning: Display file content.

10. less

```
less syslog
```

Meaning: Read large files page by page.

11. head

```
head auth.log
```

Meaning: Show first lines of file.

12. tail

```
tail auth.log  
tail -f auth.log
```

Meaning: Show last lines of file (-f = live monitoring logs).

13. grep

```
grep "Failed password" auth.log
```

Meaning: Search specific text in files.

SOC Use: Failed login detection, IOC searching, Suspicious activity finding.

14. egrep

```
egrep "error|failed|denied" syslog
```

Meaning: Advanced grep with regex.

[3. File Searching & Location]

15. find

```
find / -name "*.log"
```

Meaning: Search files/directories.

16. locate

```
locate auth.log
```

Meaning: Quickly locate files.

17. which

```
which python
```

Meaning: Show command location.

18. whereis

```
whereis ssh
```

Meaning: Find binary/source/manual of command.

[4. Permissions & Ownership]

19. chmod

```
chmod 755 script.sh
```

Meaning: Change file permissions.

20. chown

```
chown root:root file.txt
```

Meaning: Change file owner.

[5. Process Management & Monitoring]

21. ps

```
ps aux
```

Meaning: Show running processes.

SOC Use: Detect suspicious processes, Malware investigation.

22. top

```
top
```

Meaning: Real-time system monitoring.

23. htop

```
htop
```

Meaning: Interactive process viewer.

24. kill

```
kill 1234
```

Meaning: Stop process.

25. killall

```
killall firefox
```

Meaning: Kill process by name.

[6. System Performance & Resource Status]

26. df

```
df -h
```

Meaning: Disk space usage.

27. du

```
du -sh logs/
```

Meaning: Folder/file size.

28. free

```
free -h
```

Meaning: Memory usage.

29. uptime

```
uptime
```

Meaning: System uptime and load.

[7. User Auditing & Identity]

30. whoami

```
whoami
```

Meaning: Current logged-in user.

31. id

```
id
```

Meaning: User ID and groups.

32. who

```
who
```

Meaning: Logged-in users.

33. last

```
last
```

Meaning: Login history.

SOC Use: Check unauthorized logins.

34. history

```
history
```

Meaning: Previously executed commands.

SOC Use: Investigate attacker activity.

[8. Networking & Investigation Commands]

35. ping

```
ping google.com
```

Meaning: Check network connectivity.

36. ifconfig

```
ifconfig
```

Meaning: Network interface details.

37. ip

```
ip a
```

Meaning: Modern network configuration command.

38. netstat

```
netstat -antp
```

Meaning: Show network connections.

SOC Use: Detect suspicious connections.

39. ss

```
ss -tulnp
```

Meaning: Modern replacement for netstat.

40. nslookup

```
nslookup google.com
```

Meaning: DNS lookup.

41. dig

```
dig google.com
```

Meaning: Detailed DNS query tool.

42. curl

```
curl http://example.com
```

Meaning: Transfer data from URLs.

SOC Use: API testing, Malware URL checking.

43. wget

```
wget http://example.com/file.txt
```

Meaning: Download files from internet.

[9. Data Transfer & Archiving]

44. scp

```
scp file.txt user@server:/home/
```

Meaning: Secure file transfer.

45. ssh

```
ssh user@192.168.1.10
```

Meaning: Secure remote login.

46. tar

```
tar -cvf logs.tar logs/
```

Meaning: Archive/compress files.

47. unzip

```
unzip files.zip
```

Meaning: Extract zip files.

[10. Service & Log Administration]

48. journalctl

```
journalctl
journalctl -xe
```

Meaning: View systemd logs.

SOC Use: System event investigation.

49. systemctl

```
systemctl status ssh
```

Meaning: Manage services.

50. sudo

```
sudo apt update
```

Meaning: Run command as administrator.

[Most Important Commands for SOC L1]

These are heavily used in real SOC work:

Command	SOC Use
grep	Search logs
tail -f	Live log monitoring
ps aux	Suspicious process detection
netstat / ss	Network connection analysis
last	Login history
history	Investigate user actions
journalctl	System log analysis
find	Locate suspicious files
curl	Test URLs/APIs
dig / nslookup	DNS investigation

[Simple SOC Workflow Example]

```
[ Step 1: Check Failed Logins ]
└─ Command: grep "Failed password" /var/log/auth.log
    |
    ▼
[ Step 2: Check Active Connections ]
└─ Command: ss -tulnp
```

↓
[Step 3: Check Suspicious Processes]

└─ Command: ps aux



↓
[Step 4: Monitor Logs Live]

└─ Command: tail -f /var/log/syslog



↓
[Step 5: Check Login History]

└─ Command: last

[Best Commands to Master First]

1. **grep** - Log parsing and filtering master tool.
2. **tail -f** - Monitoring ongoing threats dynamically.
3. **find** - Tracking hidden scripts or file anomalies.
4. **ps aux** - Inspecting active background malware traces.
5. **ss** - Socket auditing for rapid connection sweeps.
6. **journalctl** - Direct systemd framework log inspection.
7. **history** - Reconstructing the path of attacker input execution.
8. **curl** - Probing external network entities or infrastructure.
9. **dig** - Resolving structural domain indicators.
10. **chmod** - Analyzing system files and script elevation permissions.